The Unit Circle and Exact Values

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Points on the unit circle**

The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle.

Find:

a $\sin\theta = \frac{4}{5}$

b $\cos\theta = -\frac{3}{5}$

c $\tan\theta = -\frac{4}{3}$

d the quadrant containing θ Quadrant II

(a) $= \frac{4}{5}$; (b) $= -\frac{3}{5}$; (c) $= -\frac{4}{3}$; (d) Quadrant II

2 **[Foundational]** • **Target: Points on the unit circle**

State the reference angle for each angle:

a $\frac{5\pi}{6} = \frac{\pi}{6}$

b $\frac{4\pi}{3} = \frac{\pi}{3}$

c $-\frac{\pi}{4} = \frac{\pi}{4}$

d $\frac{7\pi}{2} = \frac{\pi}{2}$

(a) $= \frac{\pi}{6}$; (b) $= \frac{\pi}{3}$; (c) $= \frac{\pi}{4}$; (d) $= \frac{\pi}{2}$

3 **[Foundational]** • **Target: Points on the unit circle**

Identify the quadrant(s) in which each pair of conditions holds:

a $\sin\theta < 0$ and $\cos\theta > 0$ Quadrant IV.

b $\tan\theta > 0$ and $\sin\theta < 0$ Quadrant III.

(a) Quadrant IV.; (b) Quadrant III.

4 **[Foundational]** • **Target: Points on the unit circle**

Given that $\cos\theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin\theta$ and $\tan\theta$ exactly.

$\sin^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$, and in QIII $\sin\theta < 0$: $\sin\theta = -\frac{12}{13}$. $\tan\theta = \frac{-12/13}{-5/13} = \frac{12}{5}$.

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5 [Foundational] • Target: Even/odd properties and periodicity

Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin\theta \ = 0$

b $\cos(-\theta) - \cos\theta \ = 0$

c $\frac{\sin(-\theta)}{\cos(-\theta)} \ = -\tan\theta$

(a) $= 0$; (b) $= 0$; (c) $= -\tan\theta$

6 [Foundational] • Target: Even/odd properties and periodicity

Use periodicity to find the exact value of each expression:

a $\sin\frac{13\pi}{6}$ $\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$: $= \frac{1}{2}$ **b** $\cos\frac{17\pi}{4}$ $\frac{17\pi}{4} = 4\pi + \frac{\pi}{4}$: $= \frac{\sqrt{2}}{2}$ **c** $\tan\frac{9\pi}{4}$ $\frac{9\pi}{4} = 2\pi + \frac{\pi}{4}$: $= 1$

(a) $\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$: $= \frac{1}{2}$; (b) $\frac{17\pi}{4} = 4\pi + \frac{\pi}{4}$: $= \frac{\sqrt{2}}{2}$; (c) $\frac{9\pi}{4} = 2\pi + \frac{\pi}{4}$: $= 1$

7 [Foundational] • Target: Solving basic trigonometric equations on the unit circle

Find all solutions to $\sin\theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.

Reference angle $\frac{\pi}{6}$, sine negative in QIII and QIV: $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Building the unit circle from special triangles

Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at $\theta = \frac{\pi}{3}$.

A 30-60-90 triangle with hypotenuse 1 has legs $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$. At $\theta = \frac{\pi}{3}$ $(60°)$, $x = \cos\frac{\pi}{3} = \frac{1}{2}$, $y = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$. Point: $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

9 [Intermediate] • Target: Building the unit circle from special triangles

State the coordinates of all four points on the unit circle where the terminal side lies along an axis ($\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$).

$\theta = 0$: $(1, 0)$. $\theta = \frac{\pi}{2}$: $(0, 1)$. $\theta = \pi$: $(-1, 0)$. $\theta = \frac{3\pi}{2}$: $(0, -1)$.

10 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

A point on the unit circle has x-coordinate $\frac{7}{25}$ and lies in Quadrant IV. Find its y-coordinate using $x^2 + y^2 = 1$.

$y^2 = 1 - \frac{49}{625} = \frac{576}{625}$, and in QIV $y < 0$: $y = -\frac{24}{25}$.

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11 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

Given $\sin\theta = 0.6$ and θ in Quadrant II, find $\cos\theta$ and $\tan\theta$ exactly (using the fact $0.6 = \frac{3}{5}$).

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, and in QII $\cos\theta < 0$: $\cos\theta = -\frac{4}{5}$. $\tan\theta = \frac{3/5}{-4/5} = -\frac{3}{4}$.

12 [Intermediate] • Target: The unit circle picture

The unit circle $x^2 + y^2 = 1$ is shown. Use it to estimate $\cos\left(\frac{2\pi}{3}\right)$ and $\sin\left(\frac{2\pi}{3}\right)$ visually, then state the exact values.

The angle $\frac{2\pi}{3}$ $(120°)$ lies in QII, roughly two-thirds around from 0. Exact values: $\cos\frac{2\pi}{3} = -\frac{1}{2}$, $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$.

13 [Intermediate] • Target: All six trig functions from a unit circle point

A point on the unit circle has coordinates $\left(-\frac{5}{13}, \frac{12}{13}\right)$. Find all six trigonometric function values for the corresponding angle θ.

$\sin\theta = \frac{12}{13}$, $\cos\theta = -\frac{5}{13}$, $\tan\theta = -\frac{12}{5}$, $\csc\theta = \frac{13}{12}$, $\sec\theta = -\frac{13}{5}$, $\cot\theta = -\frac{5}{12}$.

14 [Intermediate] • Target: Solving for the angle given exact coordinates

Find all values of θ in $[0, 2\pi)$ for which the unit circle point is exactly $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

This point lies in QIII where both coordinates are negative with reference angle $\frac{\pi}{4}$: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$ (only one solution in $[0, 2\pi)$ since the exact point pins down a single angle).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 7.04 Applications: Growth and Decay

Level 1 · FoundationsFind the exact value of $\tan\left(\frac{11\pi}{6}\right)$ and $\sec\left(\frac{11\pi}{6}\right)$ using reference angles and unit circle coordinates.

Quadrant & Reference Angle: $\frac{11\pi}{6}$ lies in Quadrant IV because $\frac{3\pi}{2}$\theta_{\text{ref}} = 2\pi - \frac{11\pi}{6} = \frac{\pi}{6}.x and \(y $\cos\left(\frac{11\pi}{6}\right) = +\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$, $\sin\left(\frac{11\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}$. Calculate Ratios: $\tan\left(\frac{11\pi}{6}\right) = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$. .

16 [Challenging] • Target: The unit circle picture • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: LHS $= \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$:
$= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x =$ RHS. ✓

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17 [Challenging] • Target: All six trig functions from a unit circle point • Cross-Lesson: 8.02 Graphing Rational Functions

Level 3 · AdvancedFind all values of θ in $[0, 2\pi)$ for which the rose curve $r = 4\sin(3\theta)$ passes through the pole (origin), and find the maximum distance of any point on the curve from the origin.

Step 1 (Find Pole Crossings where r = 0): $4\sin(3\theta) = 0 \Rightarrow \sin(3\theta) = 0$. For $\theta \in [0, 2\pi)$, the argument $3\theta \in [0, 6\pi)$. The sine function equals zero at integer multiples of π: $3\theta = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$. $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$. Step 2 (Find Maximum Distance from Pole): Distance from pole is $|r| = |4\sin(3\theta)|$. The maximum value of $|\sin(3\theta)|$ is 1. Therefore, the maximum distance from t.

18 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 8.03 Introduction to Limits

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

19 [Challenging] • Target: Exact values • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi+2\pi k}{4}\right) + i\sin\left(\frac{\pi+2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right]$, $k \in \{0, 1, 2, 3\}$. Step 3: Evaluate for each k:$k = 0$.

20 [Challenging] • Target: Points on the unit circle • Cross-Lesson: 9.02 Geometric Sequences and Series

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exact values

A point P is on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. The coordinates of P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. If a point Q on the unit circle has an angle ϕ such that $\phi = \theta + \frac{\pi}{3}$, what is the y-coordinate of point Q?

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. 1

d. $\frac{\sqrt{2}}{2}$

The coordinates of point P on the unit circle are given as $(\cos\theta, \sin\theta)$. Therefore, $\cos\theta = \frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$. This implies that $\theta = \frac{\pi}{3}$ (or 60 degrees). The point Q on the unit circle has an angle $\phi = \theta + \frac{\pi}{3}$. Substituting the value of θ, we get $\phi = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$ (or 120 degrees). The coordinates of Q are $(\cos\phi, \sin\phi)$. We need to find.

22 [Foundational] • Target: Points on the unit circle

The point P is located on the unit circle at an angle of $\frac{\pi}{3}$ radians counterclockwise from the positive x-axis. What are the Cartesian coordinates (x, y) of point P?

a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

The unit circle is defined by the equation $x^2 + y^2 = 1$. For a point on the unit circle at an angle θ from the positive x-axis, the Cartesian coordinates are given by $x = \cos(\theta)$ and $y = \sin(\theta)$. In this case, the angle is $\theta = \frac{\pi}{3}$ radians. We know that $\cos(\frac{\pi}{3}) = \frac{1}{2}$ and $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. Therefore, the Cartesian coordinates of point P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

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23 [Foundational] • Target: Even/odd properties and periodicity

A point P is located on the unit circle such that its angle with the positive x-axis is $\frac{2\pi}{3}$ radians. What are the coordinates of point P?

a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ✓

c. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{2\pi}{3}$ radians. We know that $\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$ and $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$. Therefore, the coordinates of point P are $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

24 [Foundational] • Target: Solving basic trigonometric equations on the unit circle

A circle is centered at the origin, and a point P lies on the circle. The angle formed by the positive x-axis and the line segment OP (where O is the origin) is θ. The coordinates of point P are $(r\cos\theta, r\sin\theta)$. If point P has coordinates $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and lies on a circle of radius R, find the value of R.

a. $\frac{5}{2}$

b. 5 ✓

c. $\frac{5\sqrt{3}}{2}$

d. 10

The coordinates of a point P on a circle of radius R centered at the origin are given by $(R\cos\theta, R\sin\theta)$. We are given that the coordinates of P are $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$. Therefore, we can set up the following equations: $R\cos\theta = \frac{5}{2}$ $R\sin\theta = \frac{5\sqrt{3}}{2}$ To find R, we can square both equations and add them together, using the identity $\cos^2\theta + \sin^2\theta = 1$.

25 [Foundational] • Target: Building the unit circle from special triangles

A point P lies on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume θ is in the first quadrant.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $\frac{1}{2}$

d. $\sqrt{3}$

For a point P on the unit circle, its coordinates are given by $(\cos\theta, \sin\theta)$. We are given that the x-coordinate of P is $\frac{1}{2}$, so $\cos\theta = \frac{1}{2}$. Since P is on the unit circle, its coordinates satisfy the equation $x^2 + y^2 = 1$. Substituting the x-coordinate, we get $(\frac{1}{2})^2 + y^2 = 1$. This simplifies to $\frac{1}{4} + y^2 = 1$. Subtracting $\frac{1}{4}$ from both sides gives $y^2 = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

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26 [Foundational] • Target: Using the Pythagorean identity with unit circle points

If $\sin\theta = \frac{3}{5}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{37}{50}$

b. $\frac{16}{25}$ ✓

c. $\frac{4}{5}$

d. $\frac{9}{25}$

$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$.

27 [Foundational] • Target: The unit circle picture

A point P has polar coordinates (r, θ). When converted to Cartesian coordinates, P is $(3, 3\sqrt{3})$. If the angle θ is restricted to the interval $[0, 2\pi)$, find the value of r and θ in radians.

a. $r = 6, \theta = \frac{\pi}{3}$ ✓

b. $r = 3\sqrt{2}, \theta = \frac{\pi}{4}$

c. $r = 6, \theta = \frac{\pi}{6}$

d. $r = 3, \theta = \frac{\pi}{3}$

The conversion formulas from polar coordinates (r, θ) to Cartesian coordinates (x, y) are $x = r\cos\theta$ and $y = r\sin\theta$. We are given the Cartesian coordinates $x = 3$ and $y = 3\sqrt{3}$. We can find r using the relationship $r^2 = x^2 + y^2$. $r^2 = 3^2 + (3\sqrt{3})^2$ $r^2 = 9 + (9 \times 3)$ $r^2 = 9 + 27$ $r^2 = 36$ Since r represents a distance, it must be non-negative. Thus, $r = \sqrt{36} = 6$. Now we find θ using the relationships $\tan\theta = \frac{y}{x}$ (if $xeq0$).

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28 [Foundational] • Target: All six trig functions from a unit circle point

A particle's position in the Cartesian plane is described by the polar coordinates (r, θ), where r is the distance from the origin and θ is the angle measured counterclockwise from the positive x-axis. The particle begins at $\theta_1 = \frac{\pi}{6}$ with a radial distance of $r_1 = 4$. It then moves to a new position such that its new angle is $\theta_2 = \frac{2\pi}{3}$ and the radial distance is $r_2 = 6$. If the displacement vector of the particle is $\mathbf{d} = \langle x, y \rangle$, what is the value of $x^2 + y^2$?

a. 13

b. 25

c. 49 ✓

d. 52

The initial position of the particle in Cartesian coordinates is $(x_1, y_1) = (r_1 \cos\theta_1, r_1 \sin\theta_1)$. Given $r_1 = 4$ and $\theta_1 = \frac{\pi}{6}$, we have: $x_1 = 4\cos(\frac{\pi}{6}) = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3}$ and $y_1 = 4\sin(\frac{\pi}{6}) = 4 \cdot \frac{1}{2} = 2$. So, the initial position is $(2\sqrt{3}, 2)$. The final position of the particle in Cartesian coordinates is $(x_2, y_2) = (r_2 \cos\theta_2, r_2 \sin\theta_2)$. Given $r_2 = 6$ and $\theta_2 = \frac{2\pi}{3}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving for the angle given exact coordinates

Consider a right-angled triangle ABC, where the right angle is at vertex B. Point D lies on the hypotenuse AC such that BD is perpendicular to AC. If AB = 10 units and BC = 24 units, find the length of AD.

a. $\frac{100}{26}$ ✓

b. $\frac{240}{26}$

c. $\frac{576}{26}$

d. $\frac{600}{26}$

Let the triangle be ABC with $\angle B = 90^\circ$. $BD \perp AC$, with D on AC. By the Pythagorean theorem in ΔABC, we have $AC^2 = AB^2 + BC^2$. Given $AB = 10$ and $BC = 24$, we get $AC^2 = 10^2 + 24^2 = 100 + 576 = 676$. So, $AC = \sqrt{676} = 26$ units. In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. $\Delta ABC \sim \Delta ADB \sim \Delta BDC$. From the similarity $\Delta ABC \sim \Delta ADB$, we have the ratio of corresponding sides.

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30 [Intermediate] • Target: Exact values

The terminal ray of an angle θ in standard position in the xy-plane passes through the point $(-5, 12)$. What is the value of $\sin\theta$?

a. $-\frac{5}{13}$

b. $\frac{12}{13}$ ✓

c. $-\frac{12}{5}$

d. $\frac{12}{5}$

For an angle θ in standard position whose terminal ray passes through a point (x, y), the trigonometric functions are defined using the distance from the origin to the point, $r = \sqrt{x^2 + y^2}$. 1. Calculate r: $r = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. Compute $\sin\theta$: $\sin\theta = \frac{y}{r} = \frac{12}{13}$ Common misconceptions: - $-\frac{5}{13}$ is $\cos\theta = \frac{x}{r}$. - $-\frac{12}{5}$ is $\tan\theta = \frac{y}{x}$. - $\frac{12}{5}$ incorrectly uses.

31 [Intermediate] • Target: Points on the unit circle

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓ ✓

Applying Trigonometric Functions principles, Standard derivation. Therefore, the verified correct option is $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓.

32 [Intermediate] • Target: Even/odd properties and periodicity

Pythagorean identity: $1 + \tan^2\theta =$

a. $\csc^2\theta$

b. $\sec^2\theta$ ✓

c. $\cot^2\theta$

d. $\cos^2\theta$

Applying Unit 3 · Trigonometric and Polar Functions principles, $\sec^2\theta = 1 + \tan^2\theta$ Therefore, the verified correct option is $\sec^2\theta$.

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33 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

A circle has radius 5. Find the arc length subtended by a central angle of $\frac{1\pi}{2}$ radians.

a. $5/4\pi$

b. $5/2\pi$ ✓

c. $7/2\pi$

d. 5π

$s = r\theta = 5 \cdot \frac{1\pi}{2} = 5/2\pi$.

34 [Intermediate] • Target: Building the unit circle from special triangles

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. **5** ✓

b. 7

c. 25

d. 16

The complex number can be written as $z = 3 + 4i$. The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. In this case, $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

35 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

In polar form, which equation represents a circle of radius 3 centred at the origin?

a. $r = 3$ ✓

b. $r = 3\theta$

c. $r = 3\cos\theta$

d. $\theta = 3$

r = 3 means every point is exactly 3 units from the origin — a circle of radius 3.

36 [Intermediate] • Target: The unit circle picture

Find θ (in $[0^\circ, 360^\circ)$) for the Cartesian point $(3, 0)$.

a. 180

b. **0** ✓

c. 181

d. 90

The point $(3, 0)$ lies at angle $\theta = 0^\circ$ from the positive x-axis.

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37 [Intermediate] • Target: All six trig functions from a unit circle point

Find r for the Cartesian point $(3, 4)$.

a. 7

b. 3

c. 4

d. **5** ✓

$r = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 8.03 Introduction to Limits

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2), \quad L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2).$ The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$ ✓

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

1. **Determine the positions of P and Q:** Let P on L_1 have coordinates $(1 + \lambda, 2\lambda, -1 + 2\lambda)$ and Q on L_2 have coordinates $(2 + 2\mu, -4 + \mu, 7 - 2\mu)$. The displacement vector $\vec{PQ}$ is:

$\vec{PQ} = ((2 + 2\mu) - (1 + \lambda); (-4 + \mu) - 2\lambda; (7 - 2\mu) - (-1 + 2\lambda)) = (1 + 2\mu - \lambda; -4 + \mu - 2\lambda; 8 - 2\mu - 2\lambda).$ Since $PQ \perp L_1$.

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39 [Challenging] • Target: Exact values • Cross-Lesson: 9.01 Arithmetic Sequences and Series

A curve passes through the point $(1, \frac{1}{2})$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y^2(1-\ln x)}{x^2}$ for $x > 0$. Determine the exact coordinates of the stationary point of the curve and its nature.

a. $(e, \frac{e}{2e-1})$**, which is a local maximum ✓**

b. $(e, \frac{1}{2e-1})$, which is a local maximum

c. $(e, \frac{e}{2e+1})$, which is a local minimum

d. $(\frac{1}{e}, \frac{1}{2e+1})$, which is a local minimum

1. **Separate the variables:** $\frac{1}{y^2}\,dy = \frac{1-\ln x}{x^2}\,dx$ 2. **Integrate both sides:** The left-hand side integrates to: $\int y^{-2}\,dy = -\frac{1}{y}$ For the right-hand side, split into two integrals: $\int \frac{1-\ln x}{x^2}\,dx = \int x^{-2}\,dx - \int x^{-2}\ln x\,dx$ We have $\int x^{-2}\,dx = -\frac{1}{x}$. Using integration by parts for $\int x^{-2}\ln x\,dx$ with $u = \ln x$ and $v' = x^{-2}$.

40 [Challenging] • Target: Points on the unit circle • Cross-Lesson: 9.02 Geometric Sequences and Series

Consider the exponential function $f(x) = a \cdot b^x$ where a and b are positive constants. The graph of $y = f(x)$ passes through the points $P = (1, 6)$ and $Q = (3, 54)$. A related function $g(x)$ is defined such that $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. If the point R on the graph of $y = g(x)$ has an x-coordinate of 2, what is the y-coordinate of R?

a. 2 ✓

b. 3

c. $\log_3 9$

d. $\frac{1}{2}\log_3 18$

The function is given by $f(x) = a \cdot b^x$. It passes through $(1, 6)$ and $(3, 54)$. \\ We have two equations: \\ 1) $6 = a \cdot b^1$ \\ 2) x \\ $9 = b^2$ \\ Since b is a positive constant, $b = 3$. \\ Substitute $b = 3$ into equation (1): \\ $6 = a \cdot 3$ \\ $a = 2$ \\ So, the function is $f(x) = 2 \cdot 3^x$. \\ The related function is $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. Substituting $a = 2$ and $b = 3$, we get: \\ $g(x) = \log_3\left(\frac{2 \cdot 3^x}{2}\right)$ \\ $g(x) = \log_3(3^x)$ \\ Us.

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41 [Challenging] • Target: Even/odd properties and periodicity • Cross-Lesson: 9.03 The Binomial Theorem

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

a. $\frac{12}{13}$ ✓

b. $-\frac{5}{13}$

c. $-\frac{12}{13}$

d. $-\frac{12}{5}$

To find $\sin(\theta)$ for an angle in standard position whose terminal side passes through $(x, y) = (-5, 12)$: 1. Calculate the distance r from the origin to the point: $r = \sqrt{x^2 + y^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. By the coordinate definition of trigonometric functions: $\sin(\theta) = \frac{y}{r} = \frac{12}{13}$ Misconception checks: - $-\frac{5}{13}$ results from confusing sine with cosine, where $\cos(\theta) = \frac{x}{r}$. - $-\frac{12}{13}$ comes from mistakenly as.

42 [Challenging] • Target: Solving basic trigonometric equations on the unit circle • Cross-Lesson: 1.01 Functions and Function Notation

The graph of the function $f(x) = A\sin(Bx + C) + D$ is shown below. The maximum value of the function is 5 and the minimum value is -1. The period of the function is $\frac{\pi}{2}$, and the graph passes through the point $(0, 2)$. Find the value of A.

a. 2

b. **3** ✓

c. 4

d. 6

The amplitude A of a sinusoidal function is half the difference between its maximum and minimum values. \\ Given: Maximum value = 5, Minimum value = -1 \\ Amplitude x is 3. \\ The other parameters (period and phase shift) can be used to find B, C, and D, but are not necessary to determine A.

43 [Challenging] • Target: Building the unit circle from special triangles • Cross-Lesson: 1.02 Domain and Range

Find all values of θ such that $\sin\theta, \cos\theta, \tan\theta$ are in arithmetic progression.

a. $\theta \approx 0.6435$ **rad (a special angle satisfying** $2\cos\theta = \sin\theta + \tan\theta$**)** ✓

b. $\theta = \pi/3$

c. No solution exists

d. $\theta = \pi/4$

$2\cos\theta = \sin\theta + \tan\theta \Rightarrow 2\cos^2\theta\sin\theta = \sin^2\theta + \sin^2\theta/\cos\theta \cdot \cos\theta$. Complex transcendental equation.

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44 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 1.03 Average Rate of Change

State the Pythagorean identity relating $\sec$ and $\tan$.

a. $\sec^2 x = \tan^2 x$

b. $\tan^2 x + 1 = \csc^2 x$

c. $\sec^2 x = 1 + \tan^2 x$ ✓

d. $\sec^2 x = 1 - \tan^2 x$

Applying Trigonometric Functions principles, $1 + \tan^2 x = \sec^2 x$. Therefore, the verified correct option is $\sec^2 x = 1 + \tan^2 x$.

45 [Challenging] • Target: The unit circle picture • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Convert the rectangular point $(-1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

a. $r = 1.414,\ \theta = 225°$

b. $r = 2.414,\ \theta = 135°$

c. $r = 1.414,\ \theta = 315°$

d. $r = 1.414,\ \theta = 135°$ ✓

$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (1)^2} \approx 1.414$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 135°$.

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